Acknowledgments

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Appendix

Appendix A. Proof of Theorem 4.1

The proof of Theorem 4.1 relies on standard arguments from dynamical systems, upon noticing that the evolution (4.1) is a (continuous-time) gradient ascent for the energy $\mathsf{E}_0 : (\mathbb{S}^{d-1})^n \to \mathbb{R}$ defined as

$$\mathsf{E}_0(x_1, \dots, x_n) = \frac{1}{n} \sum_{i=1}^{n} \sum_{j=1}^{n} \langle x_i, x_j \rangle.$$

Since the dynamics are the gradient ascent of a real-analytic functional on the compact real-analytic manifold $(\mathbb{S}^{d-1})^n$, the celebrated Łojasiewicz theorem [Loj63], in the form given by [HKR18, Corollary 5.1]—which is valid in the context of general compact Riemannian manifolds—, implies that for any initial condition $X \in (\mathbb{S}^{d-1})^n$, the solution $\Phi^t(X) \in (\mathbb{S}^{d-1})^n$ converges to some critical point $X^* \in (\mathbb{S}^{d-1})^n$ of E_0 as $t \to +\infty$.

We recall that a *strict saddle point* of E_0 is a critical point of E_0 at which the Hessian of E_0 has at least one strictly positive eigenvalue. Theorem 4.1 then follows by combining the following couple of lemmas with the Łojasiewicz theorem.

Lemma A.1. *Let $\mathcal{M}$ be a compact Riemannian manifold and let $f : \mathcal{M} \to \mathbb{R}$ be a smooth function. The set of initial conditions $X_0 \in \mathcal{M}$ for which the gradient ascent*

$$\begin{cases} \dot{X}(t) = \nabla f(X(t)) \\ X(0) = X_0 \end{cases} \tag{A.1}$$

converges to a strict saddle point of f is of volume zero.

Proof of Lemma A.1. Let us denote by $\Phi^t(X_0) := X(t)$, $t \geqslant 0$ the solution to (A.1). We denote by $\mathcal{S} \subset \mathcal{M}$ the set of strict saddle points of f, and by $\mathcal{A} \subset \mathcal{M}$ the set of initial conditions $X_0 \in \mathcal{M}$ for which $\Phi^t(X_0)$ converges to a strict saddle point of f as $t \to +\infty$. For any $y \in \mathcal{S}$, we denote by B_y a ball in which the local center-stable manifold $W^{\mathrm{sc}}_{\mathrm{loc}}(y)$ exists (see [Shu13], Theorem III.7 and Exercise III.3 for the adaptation to flows). Using compactness, we may write the union of these balls as a countable union $\bigcup_{k \in I} B_{y_k}$ (where I is countable and $y_k \in \mathcal{M}$ for $k \in I$). If $X_0 \in \mathcal{A}$, there exists some $t_0 \geqslant 0$ and $k \in I$ such that $\Phi^t(X_0) \in B_{y_k}$ for all $t \geqslant t_0$. From the center-stable manifold theorem ([Shu13], Theorem III.7 and Exercise III.3, where we note that the Jacobian of a gradient vector field coincides, at a critical point, with the Hessian of the corresponding function) we gather that $\Phi^t(X_0) \in W^{\mathrm{sc}}_{\mathrm{loc}}(y_k)$

for $t \geqslant t_0$, hence $X_0 \in \Phi^{-t}(W^{\mathrm{sc}}_{\mathrm{loc}}(y_k))$ for all $t \geqslant t_0$. The dimension of $W^{\mathrm{sc}}_{\mathrm{loc}}(y_k)$ is at most $\dim(\mathcal{M}) - 1$, thus it has zero volume. Since Φ^t is a diffeomorphism on a compact manifold, Φ^{-t} preserves null-sets and hence $\Phi^{-t}(W^{\mathrm{sc}}_{\mathrm{loc}}(y_k))$ has zero volume for all $t \geqslant 0$. Therefore $\mathscr{A}$, which satisfies

$$\mathscr{A} \subset \bigcup_{k \in I} \bigcup_{\ell \in \mathbb{N}} \Phi^{-\ell}(W^{\mathrm{sc}}_{\mathrm{loc}}(y_k))$$

has volume zero. □

Lemma A.2. *Any critical point $(x_1, \ldots, x_n) \in (\mathbb{S}^{d-1})^n$ of E_0 which is not a global maximum, namely such that $x_1 = \ldots = x_n$, is a strict saddle point. In particular, all local maxima are global.*

Proof of Lemma A.2. We extend the proof idea of [Tay12, Theorem 4.1] as follows. Let $(x_1, \ldots, x_n) \in (\mathbb{S}^{d-1})^n$ be a critical point of E_0, and assume that the points x_i are not all equal to each other.

Step 1. We first prove that there exists a set of indices $\mathcal{S} \subset [n]$ such that

$$\sum_{i \in \mathcal{S}} \sum_{j \in \mathcal{S}^c} \langle x_i, x_j \rangle < 0. \tag{A.2}$$

To this end, define

$$m := \sum_{j=1}^{n} x_j,$$

and consider two cases. If $m \neq 0$, then we deduce from $\nabla \mathsf{E}_0(x_1, \ldots, x_n) = 0$ that for any $j \in [n]$, x_j is collinear with m. Thus $x_j = \pm x_1$ for any $j \in [n]$. Setting

$$\mathcal{S} = \{j \in [n] \colon x_j = +x_1\},$$

we can see that (A.2) holds, unless $\mathcal{S} = [n]$ which has been excluded. Now suppose that $m = 0$. Then by expanding $\langle m, x_i \rangle = 0$, we find that for any $i \in [n]$

$$-1 = \sum_{j=2}^{n} \langle x_j, x_1 \rangle,$$

holds, which again implies (A.2) with $\mathcal{S} = \{1\}$.

Step 2. In this second step we look to deduce from (A.2) that $(x_1, \ldots, x_n)$ is a strict saddle point. Consider an arbitrary non-zero skew-symmetric matrix B and define the perturbation

$$x_i(t) = \begin{cases} x_i & i \notin \mathcal{S} \\ e^{tB} x_i & i \in \mathcal{S}. \end{cases}$$

Set $\mathsf{E}_0(t) = \mathsf{E}_0(x_1(t), \ldots, x_n(t))$. Note that we have

$$\mathsf{E}_0(t) = \text{const.} + \frac{2}{n} \sum_{i \in \mathcal{S}} \sum_{j \in \mathcal{S}^c} \langle x_i(t), x_j \rangle,$$

where we grouped time-independent terms into the constant (recall that e^{tB} is an orthogonal matrix, since skew-symmetric matrices are the Lie algebra of $\mathrm{SO}(d)$).

Thus

$$\mathsf{E}_0'(t) = \frac{2}{n}\sum_{i\in\mathcal{S}}\sum_{j\in\mathcal{S}^c}\langle \dot{x}_i(t), x_j\rangle$$
$$\mathsf{E}_0''(t) = \frac{2}{n}\sum_{i\in\mathcal{S}}\sum_{j\in\mathcal{S}^c}\langle \ddot{x}_i(t), x_j\rangle\,.$$

Since $(x_1,\dots,x_n)$ is a critical point of E_0, we have $\mathsf{E}_0'(0)=0$. On the other hand, since $\ddot{x}_i(0) = B^2 x_i$ we have

$$\mathsf{E}_0''(0) = \frac{2}{n}\sum_{i\in\mathcal{S}}\sum_{j\in\mathcal{S}^c}\langle B^2 x_i, x_j\rangle\,. \tag{A.3}$$

We claim that given (A.2), there must exist some skew-symmetric matrix B such that $\mathsf{E}_0''(0) > 0$. Indeed, if d is even, then we just take B as the block-diagonal matrix with repeated block

$$\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix},$$

so that $B^2 = -I_d$. If d is odd, we can represent

$$-I_d = \frac{1}{d-1}\sum_{j=1}^{d} B_j^2, \tag{A.4}$$

where B_j is the same block-diagonal matrix, with the exception that the j-th block is a 1×1 zero-matrix. If each B_j were to yield $\mathsf{E}_0''(0) \leqslant 0$, then it would violate (A.2). Thus, $\mathsf{E}_0''(0) > 0$ for some well-chosen skew-symmetric B, which proves that $(x_1,\dots,x_n)$ is a strict saddle point. □

Appendix B. Proof of Theorem 5.1

Proof of Theorem 5.1. We leverage the gradient flow structure presented in Remark 3.8 and Section 3.4 (the manifold is compact, and the metric and functional are analytic), and use Lemma A.1 as in the proof of Theorem 4.1. Consequently, it suffices to show that, in the stated regime of β, the critical points of E_β which are not global maxima are strict saddle points, namely, that all local maxima are global. For simplicity we write the argument for (USA) and explain the extension to the case of (SA) in Remark B.1.

We begin by focusing on the case $d=2$, and provide a brief argument which shows that the case of arbitrary $d \geqslant 2$ readily follows.

Let $(\theta_1,\dots,\theta_n)\in\mathbb{T}^n$ be a critical point such that all eigenvalues of the Hessian of E_β are non-positive. We intend to show that if β is sufficiently large, then necessarily $\theta_1 = \cdots = \theta_n$. To that end, note that the non-positivity of the Hessian of E_β implies in particular that for any subset of indices $\mathcal{S}\subset[n]$, we must have

$$\sum_{i\in\mathcal{S}}\sum_{j\in\mathcal{S}}\partial_{\theta_i}\partial_{\theta_j}\mathsf{E}_\beta(\theta_1,\dots,\theta_n) \leqslant 0\,. \tag{B.1}$$

Notice that for any $i,j\in[n]$,

$$\partial_{\theta_i}\mathsf{E}_\beta(\theta_1,\dots,\theta_n) = \frac{1}{n^2}\sum_{m\in[n]\setminus\{i\}} -\sin(\theta_i-\theta_m)e^{\beta\cos(\theta_i-\theta_m)}$$